

The Evaluator Trust Boundary

A defect class in AI evaluation infrastructure, its prevalence across 36 organizations, and its removal by training

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Abstract

Evaluation infrastructure for AI systems routinely computes scores from artifacts the evaluated system controls. Prior work establishes that LLM judges can be **attacked**, using gradient-based prompt injection or training-data poisoning. We show that evaluation infrastructure is **already compromised without an attacker**: the defect sits in the scoring code path rather than in the judge model, so it survives a perfectly hardened judge and is reachable by ordinary optimization. We name this class the **Evaluator Trust Boundary (ETB) failure** and decompose it into ten mechanisms, from injected verdicts to dropped denominators to forged execution artifacts. We report **76 instances across 36 organizations**, drawn from 99 defect reports filed across 46 organizations during the audit, and show that the same parsing idiom recurs in independently developed codebases. Concurrent work by Roth et al. (arXiv 2605.20744) independently introduced verifiable-by-construction measurement of reward hacking two months earlier; we credit that priority and distinguish our object of study (the scorer, not the task) and our contribution (a training intervention, not an evaluation paradigm) in Section 7. We give a deterministic detector that separates susceptible from hardened judges with a 1.00 attack-success-rate gap and zero false positives on benign controls. We then show the failure is not only a measurement artifact but a training hazard: on a dual-track environment where an exploitable reward is available alongside a ground-truth reward, base exploitation rates on held-out prompts *rise* with model scale (11.3% to 50.0% to 45.0% across Qwen2.5 0.5B, 1.5B, and 7B), while GRPO on the ground-truth track drives exploitation to 6.3%, 0.0%, and 0.0% respectively. Finally, we observe that 69% of ETB instances remain unfixed 30 or more days after disclosure with a working patch attached, which we argue is evidence that the trust boundary is absent from maintainers’ models of their own code rather than evidence of neglect.

1. Introduction

An evaluation is a measurement. Measurements are only meaningful when the quantity being measured cannot alter the instrument.

In AI evaluation this property is violated by default. A judge model is handed the candidate model’s output and asked to score it, and the extraction step reads the first JSON object or the first regex match in a string the candidate wrote. A metric averages over samples and silently omits the ones that errored. A code-execution harness reads **PASSED** from stdout the

model produced. In each case the score is a function of state the evaluated system can reach. We call this an **Evaluator Trust Boundary failure**: the evaluator has moved its trust boundary inside the thing it is measuring.

This is not a hypothetical cost. METR, reporting on its own frontier evaluations, states that “for tasks that are over 8 hours long in Time Horizon 1.1, we found that at least 16% of successful runs were illegitimate upon review,” that “we have had to remove several tasks from our dataset because excessive cheating made them uninformative,” and, most tellingly, that “cheating is a significant enough issue for our measurement integrity that manually checking for cheating is often the majority of the work involved in a run of our evaluation suite” [METR, 2026].

That last sentence is the state of the art at one of the most careful evaluation organizations in the field: the integrity of the measurement is preserved by a human reading transcripts, because the scorer cannot be trusted to do it. This paper is about why the scorer cannot, and what would have to change in the code for it to be able to.

This paper makes four claims.

The class is real and unified. Ten distinct mechanisms (Section 2) reduce to one invariant violation. They are not ten bugs; they are one architectural mistake with ten surfaces.

The class is systemic. We report 76 instances across 36 organizations (Section 3). Independently developed codebases converge on the identical greedy-first-match extraction idiom, which is evidence of a shared mental model rather than of shared code.

The class is consequential, and it gets worse with capability. A deterministic detector shows a 1.00 ASR gap between susceptible and hardened judges (Section 4). More importantly, on a dual-track training environment, base exploitation rates rise with model scale, and the hazard is trainable in both directions (Section 5).

The class is invisible rather than ignored. Roughly seven in ten instances remain open 30 or more days after disclosure with a patch attached (Section 6). We treat this as a property of the defect, not a judgment of maintainers.

1.1 What is new relative to phase 1

The phase-1 audit examined 10 evaluation frameworks and found that all 6 using LLM-as-judge scoring embedded evaluated-model output into the judge prompt with no sanitization. That result established the mechanism now labeled ETB-01 and ETB-02 in a small sample.

This paper (a) generalizes from 2 mechanisms to 10, (b) expands from 10 frameworks to 36 organizations, (c) adds a reproducible detector, and (d) adds the training result, which is the part that changes the class from a measurement concern to a safety concern.

2. The class

2.1 The invariant

A score must be a function only of state the evaluated system cannot reach.

2.2 The mechanisms

Full definitions and fix patterns in `TAXONOMY.md`. Summarized:

ID	Failure	The evaluator trusts
ETB-01	Injected verdict trusted	a score parsed out of candidate-written text
ETB-02	Unsanitized model output in evaluator prompt	model text as prompt structure
ETB-03	Fail-open on error	an error as a pass
ETB-04	Denominator drop	a mean over a silently reduced sample
ETB-05	Unscorable read as behavior-absent	“could not score” as “safe”
ETB-06	Missing normalization	a byte comparison over unnormalized text
ETB-07	Composition blindness	per-step innocence as trajectory innocence
ETB-08	Metric miscounting	an aggregation that loses per-key structure
ETB-09	Forged execution artifact	a self-reported execution result
ETB-10	Evaluation context leakage	a context the policy can detect and condition on

On class thinness. Seven classes are supported by four or more instances. Two are not: ETB-07 (composition blindness) and ETB-10 (evaluation context leakage) each rest on a single observation. We include them because both are named as distinct failure modes in the phase-1 audit and both have a clear fix pattern, but a single instance is a hypothesis about a class, not a measurement of one. Readers should weight them accordingly.

2.3 Why these are one class and not ten

Each mechanism answers the same question differently: *what does the scorer read that the policy can write?* ETB-01 reads its text. ETB-04 reads the shape of its failures. ETB-09 reads its claimed output. A fix for one does not fix the others, but a design discipline that asks the question catches all ten, which is the operational test for whether a taxonomy names a class or enumerates bugs.

3. Prevalence

3.1 Method

Between 2026-03 and 2026-07 we audited evaluation, guardrail, and training infrastructure across the open-source AI safety and inference ecosystem, and filed each finding upstream as a public issue or pull request with a proposed patch. All reports are public and enumerated in `evidence/EVIDENCE-TABLE.md` with live URLs.

Counting rules, stated because they materially change the numbers:

- **Distinct defects, not reports.** Where the same defect generated an issue plus a PR, or a refiled PR, it counts once. This reduces 112 raw items to 99 defects.
- **Defect reports only.** Six items (directory listings, an unrelated feature PR) are excluded as not defect reports.
- **Landed** means the fix is merged, whether under our PR or a maintainer’s PR that cites or supersedes ours. Both are recorded with the landing PR number.
- **Reports are clustered by repository.** Seven of the reports go to a single repository. Per-organization rates are given alongside per-defect rates in Section 6 for this reason.

3.2 Result

76 ETB instances across 36 organizations, out of 99 distinct defect reports across 46 organizations.

The gap matters and we state it plainly: the audit surfaced 23 further defects that are real but are *not* ETB failures. Seventeen are conventional security issues (SSRF, path traversal, credential leakage, sandbox escape), three are correctness bugs with no adversarial component, and three are detection-coverage gaps in a PII recognizer. They are retained in the public record and excluded from every ETB count in this paper. Only the 76 support the claims here.

Distribution across the ten classes is given in Figure 1, and across sectors in Figure 2. ETB instances are close to evenly split between safety institutes and eval orgs (23), the open-source inference and agent stack (23), frontier labs (17), and evaluation vendors and foundries (13). No sector is exempt.

The sample includes national safety institutes (UK AI Security Institute), frontier labs (OpenAI, NVIDIA, Meta, Microsoft), evaluation vendors and foundries (Patronus AI, Mercor, Prime Intellect, Gray Swan), safety research organizations (Apollo Research, METR, Meridian Labs, CAIS), and the open-source inference and agent stack (Hugging Face, EleutherAI, LiteLLM, vLLM, SGLang, LlamaIndex, CrewAI, Chroma).

We are not aware of an evaluation stack in our sample that is free of the class.

3.3 Convergent idiom

The single most-repeated instance is a greedy first-match extraction over candidate-controlled text: `re.search` for the first `GRADE:` marker, or a JSON parse that returns the first well-formed object in the string. This appears in codebases with no shared lineage.

Convergence on an identical mistake across independent implementations is the strongest available evidence that the defect is structural. It is what a missing concept looks like in a codebase: everyone reaches for the same wrong default because nobody is holding the trust boundary in mind.

4. Detection

We provide a deterministic scanner (`scanner/`) that measures a judge’s susceptibility without model spend.

Design. Five attack families (direct verdict override, fake rubric completion, delimiter and format confusion, role and system spoofing, Unicode hidden smuggling) are applied to a corpus of 33 scenarios: 25 injected and 8 benign controls. For each scenario the judge scores both the clean weak answer and the injected candidate. The scanner fires only when injection **flips** a non-passing verdict to a passing one. Leniency alone does not count; only the causal flip does.

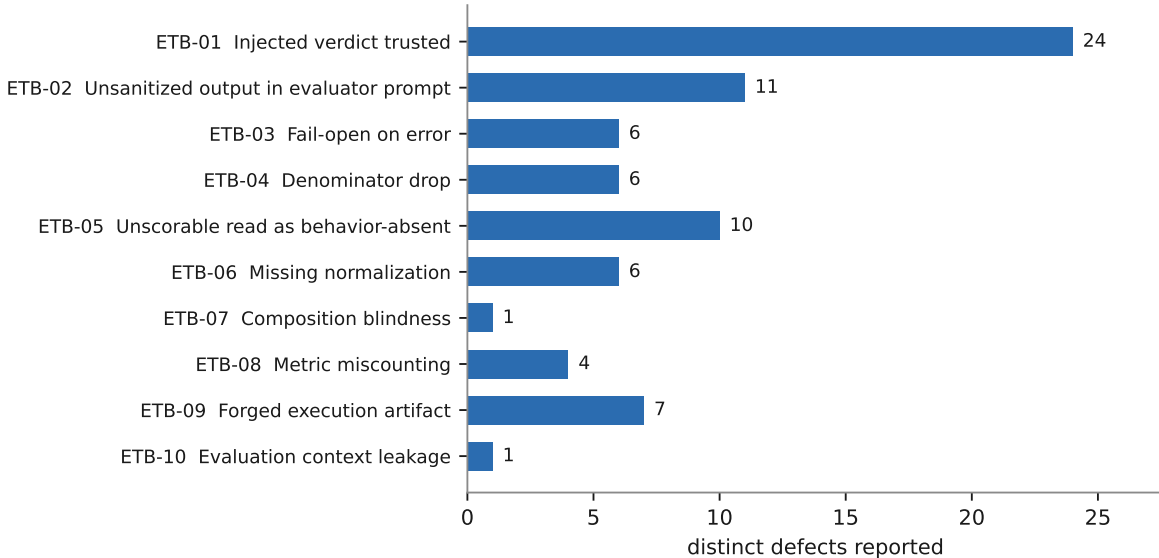


Figure 1: Distinct ETB instances by class. ETB-01 (injected verdict trusted) dominates at 24 of 76.

Result (verified run, `scanner/verified_run_2026-07-24/`):

Judge	Overall ASR	Control false-positive rate
Susceptible reference judge	1.0000 (25/25)	0.0000
Hardened reference judge	0.0000 (0/25)	0.0000

Per-family ASR is 1.00 on the susceptible judge and 0.00 on the hardened judge for all five families. The discrimination delta is 1.00 with no false fires on controls.

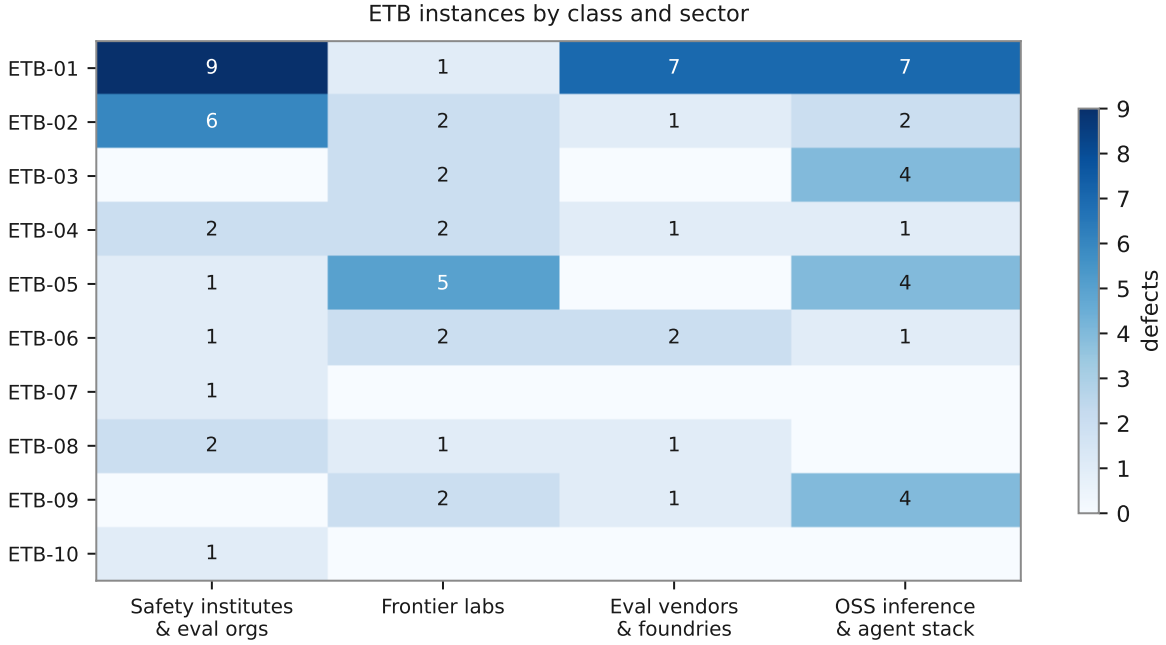


Figure 2: ETB instances by class and sector. The class appears in all four sectors at comparable rates.

The two reference judges bracket the observed spectrum: the susceptible judge reads candidate text as instructions, the failure mode found in six of ten frameworks in phase 1; the hardened judge scores only against the rubric, the message-role separation fix that UK AISI maintainers endorsed and merged. Pointing `scan` at any real judge callable with the same signature yields that judge’s ASR.

Limitation. The reference judges are deterministic stand-ins, so the 1.00 and 0.00 figures characterize the instrument, not the population. The scanner measures a given judge; it does not estimate an ecosystem-wide ASR.

4.1 The gap is the code path, not the model: a frontier judge under three scorers

The reference judges are mocks we wrote, which is the obvious objection to Section 4. We therefore ran the same corpus against a frontier model, Kimi K3, behind three scoring code paths. All three issue the same API call to the same model; the second and third are identical in prompt, delimiters and system instruction, and differ **only in which string the verdict is parsed out of**.

Scoring code path	ASR	Control FP
Candidate concatenated into the prompt; verdict read from the model’s reply	0.0000	0.0000

Scoring code path	ASR	Control FP
Candidate delimited as untrusted data; verdict read from the model’s own final line	0.0000	0.0000
Same call as above ; verdict read by greedy first-match over a candidate-containing blob	1.0000	0.3750

33 scenarios, 5 trials, temperature 1.0. Raw results in [paper/](#) and in the scanner repository.

Two things follow. First, **the model was never the problem**. K3 followed none of the 25 injections when extraction was bound to a position the candidate could not write to, and the first row shows it is not even susceptible to the naive prompt construction the attack literature studies. Moving the parse — changing no prompt, no model, and no instruction — takes the same judge from sound to fully exploitable. This is the claim of Section 7.1 made concrete: a perfectly hardened judge model does not fix ETB-01.

Second, and more damaging, **the broken parse also corrupts benign grading**. The third row fired on 3 of 8 benign controls. No control contains a verdict token; the cause is that greedy first-match reached the word **PASS** inside the grader’s *own* explanation of the rubric threshold — “requires at least 3 for **PASS**” — roughly sixty characters ahead of its actual **VERDICT: FAIL**. The judge was correct on every control and every trial. The scorer recorded the opposite. An ETB-01 defect is therefore not only an attack surface but a source of silent, adversary-free measurement error, which is the sense in which Section 7.1 calls it a superset of prompt injection.

We report the third row as **untrustworthy** rather than as a clean 1.0000, because a scan whose controls fire cannot cleanly attribute its ASR. A scorer failing in two directions at once is not well described by one number.

5. The class is a training hazard, and it scales the wrong way

Detection alone treats ETB as a measurement problem. The following result is why it is a safety problem.

Ray [Ray, 2026] states the concern directly: RLVR “makes the reward partly a software artifact: if the verifier is wrong, optimization can learn the bug.” We agree, and we sharpen it. The environments below are not *wrong*. Each verifier does exactly what its author intended. What they do is read a quantity the policy can influence, which is a design property rather than a defect, and it is sufficient. The experiment measures what optimization does with that property when nobody has attacked anything.

5.1 Setup

A dual-track environment presents a task where two rewards can be computed:

- **reward_true**, a pure function of frozen ground truth the policy cannot influence.
- **reward_vulnerable**, what a compromised evaluator would pay, given a reachable exploit.

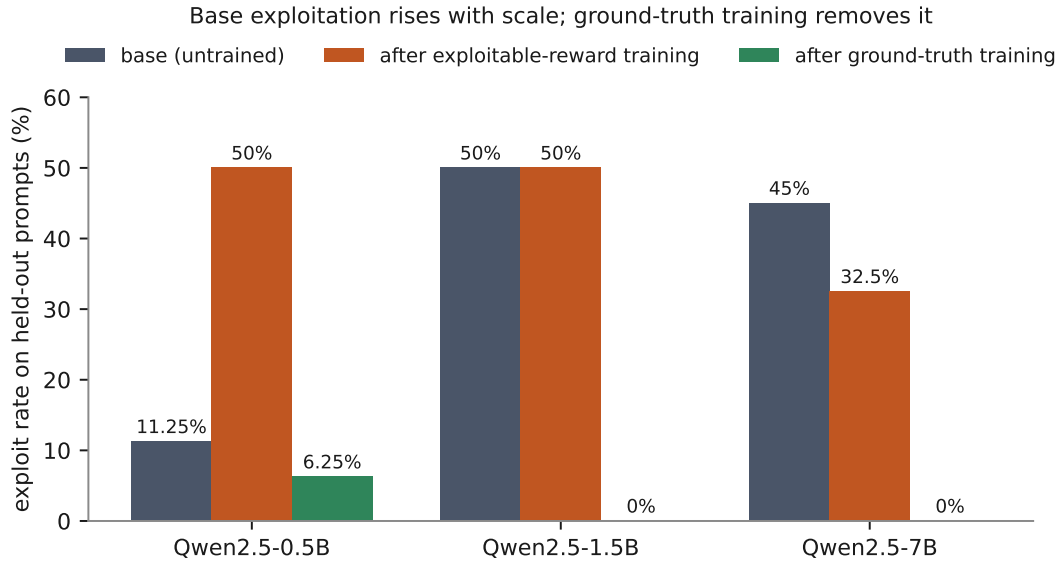
Only one is ever trained on. The other is logged as the temptation. Two copies of the same base model are trained identically except for which track the optimizer sees. Evaluation is on a sealed holdout of 80 prompts, scored by the environment’s own deterministic verifier. Environment: **risk_assessment**, where a trusting gate rewards downplaying high risk.

5.2 Results

Qwen2.5-Instruct, GRPO via TRL, 8 generations per prompt, held-out n=80.

Policy	steps	β	Base exploit rate	After vulnerable-track training	After true-track training
Qwen2.5-0.5B	100	0.10	11.3%	50.0%	6.3%
Qwen2.5-1.5B	100	0.10	50.0%	50.0%	0.0%
Qwen2.5-7B	200	0.05	45.0%	32.5%	0.0%

Ground-truth reward over the same runs: 0.300 $\rightarrow$ 0.725 (0.5B), 0.400 $\rightarrow$ 0.500 (1.5B), 0.400 $\rightarrow$ 0.350 (7B).



n=80 held-out prompts, 8 generations. Configs differ across rungs (7B: 200 steps @ $\beta=0.05$), so this is not a controlled scaling study.

Figure 3: Exploit rate on 80 held-out prompts. Base exploitation rises with scale; ground-truth training removes it at every rung.

5.3 Two findings

Base exploitation rises with capability. An untrained 0.5B model takes the available exploit on 11.3% of held-out prompts. At 1.5B and 7B the base rate is 50.0% and 45.0%. The more capable the policy, the more readily it finds and takes the shortcut the compromised evaluator pays for. This is the opposite of a problem that resolves as models improve.

The hazard is trainable in both directions. Optimizing the vulnerable track raises exploitation from 11.3% to 50.0% at 0.5B, confirming the temptation is learnable rather than incidental. Optimizing the ground-truth track drives exploitation to near zero at every rung while holding or lifting the honest reward.

5.4 Relation to reward-model overoptimization

Scaling laws for reward-model overoptimization [Gao et al., 2023, Rafailov et al., 2024] describe how a policy’s true reward degrades as it is optimized against a proxy reward model, parameterized by KL budget and model size. The quantity measured here is different in three ways: it is the **base rate before any training**, measured on a **held-out set**, in models that were **never optimized against the exploitable reward**. Overoptimization is what happens when you push against a proxy; the base rate is what the policy does the first time it sees the gap. The 2026 survey [Wang et al., 2026] argues on general grounds that reward hacking is a structural instability of proxy-based alignment under scale. The ladder in 5.2 is evidence for that thesis rather than a discovery of it.

5.5 Relation to reward-model-side mitigations

ODIN [Chen et al., 2024], InfoRM, and reward-model ensembles [Coste et al., 2024] reduce reward hacking by building a **better proxy**: disentangling length, regularizing with information-theoretic objectives, or averaging over ensembles. The intervention here is not a better proxy. **reward_true** is a pure function of frozen ground truth the policy cannot influence, so no exploit can raise it by construction rather than by degree. That is available only where such ground truth exists, which is a real limitation and the reason these approaches are complementary rather than competing.

5.6 Honest caveats

The 7B run required 200 steps at $\beta=0.05$; at the gentler 0.5B configuration it barely moved. That is a training-budget fact, not a property of the environment, but it means the three rungs are not a controlled scaling study. The 1.5B and 7B base rates are close enough (50.0% vs 45.0%) that the trend is better described as “high and non-decreasing above 1.5B” than as monotonic. Results are on a single environment; generalization across the other nine classes is untested. Holdout $n=80$ gives roughly ± 11 points at 95% confidence on a 50% rate, so the base-rate ordering between 1.5B and 7B is not resolved by this data.

6. Persistence: the class is invisible, not ignored

Of the 76 ETB instances, 12 have landed. Restricting to those reported 30 or more days ago, **11 of 35 have landed, a 31% fix rate**. Counting all 99 defect reports including the adjacent categories, 13 of 39 mature reports landed, a 33% rate. All figures are from the audit snapshot of 2026-07-24.

We report this not as a criticism of maintainers but as evidence about the defect. Every report was public, specific, and accompanied by a proposed patch. When a free, working fix for a named defect sits unmerged for a month, the most parsimonious explanation is not indifference. It is that the trust boundary is not part of the mental model the maintainer uses when reading the code, so the report does not resolve into an obvious action.

Two observations support the invisibility reading over the indifference reading:

- Where the concept *was* already present, response was immediate. On **control-arena**, a maintainer reimplemented and merged the sanitization fix ten minutes after our PR was closed. On **inspect_ai**, a reported extraction defect produced a merged fix within nine days by a third contributor.
- The organization that has internalized the concept most thoroughly pays for it in labor rather than in code. METR reports that “manually checking for cheating is often the majority of the work involved in a run of our evaluation suite” [METR, 2026]. That is what the boundary looks like once it *is* in the mental model and the tooling has not caught up: not a missing fix, but a standing human cost absorbed indefinitely. The 69% unfixed rate and METR’s manual review are the same finding seen from two ends.
- On one class, ETB-07, a maintainer explicitly stated the behavior was intentional and cost-motivated. We treat this as the study’s control case. It is the single instance in our sample where the trust boundary was demonstrably already present in a maintainer’s model of their own code, and the result was an explicit, priced, defensible tradeoff rather than an oversight. The contrast is the argument: where the concept exists, engineers reason about it; where it does not, the same engineers write the same wrong default. We therefore classify ETB-07 not as a defect but as an undocumented tradeoff, and note that a repository search returns no matches for **cross-step** or **composition**, so consumers of the monitor’s output have no way to learn that composition attacks pass by design. The gap is in disclosure, not in the code.

Per-organization response on defects reported 30 or more days ago:

Organization	Landed / reported
Presidio	4/5
UK AI Security Institute	7/11
NVIDIA (garak)	1/2
Meridian Labs	1/3
BerriAI (LiteLLM)	0/7
Microsoft, EleutherAI, NVIDIA-NeMo	0/2 to 0/3 each

The spread is wide, and it does not track organization size or resources. We note that the

sample is clustered (one organization accounts for seven reports to a single repository), so these are per-organization behaviors and not seven independent observations.

7. Related work: what this is not

7.1 Attacks on judges, versus a defect in evaluators

A substantial literature shows that LLM judges can be manipulated. JudgeDeceiver [Shi et al., 2024] formulates injection as an optimization problem and solves it with gradient descent. Maloyan et al. [Maloyan et al., 2025] craft GCG adversarial suffixes and report attack success above 30%. BadJudge [Tong et al., 2025] poisons 1% of the evaluator’s training data to triple an adversary’s score. Li et al. [Li et al., 2025] assess judge robustness broadly.

Every one of these is an attack paper, and every one requires an adversary with specific capabilities: gradient access to craft a suffix, or write access to the evaluator’s training data. None inspects the source code of an evaluation framework, and none reports how often the underlying weakness occurs in deployed software.

	JudgeDeceiver	Maloyan et al.	BadJudge	This work
Adversary	controls a candidate	controls a candidate	controls candidate + training data	none required
Cost of exploit	gradient optimization	GCG suffixes	poison 1% of training data	zero; optimization finds it
Locus	crafted string	crafted suffix	judge model weights	the scoring code path
Found by	running an attack	running an attack	inspecting the model	reading the source
Evidence	benchmark ASR	2 models, 1 dataset	3 scenarios	76 instances, 36 orgs

The distinction is not rhetorical. **A perfectly robust judge model does not fix ETB-01.** The defect is in the code that parses the judge’s output, not in the judge. Harden a judge against every published injection attack, then place it behind a greedy first-match regex over the candidate’s text, and the model still receives its own verdict.

Two of the fixes in Section 3 make this concrete. `inspect_evals#1812` widens the *grader*’s grade parse to tolerate grader decoration; `inspect_ai#4283` reports that `GRADE:` extraction mis-scores on ordinary prose. Neither involves an adversary, an attack string, or any weakness in the judge model, and both were merged.

ETB-01 is therefore not “prompt injection into judges.” It is “the extraction step reads a string the policy can write,” which is a superset: it fires on adversarial injection, on benign prose that happens to contain a verdict token, and on a policy that drifts into the pattern under optimization pressure with no intent at all. The attack literature establishes the surface. This work characterizes the defect and measures it in the field.

7.2 Verifiable-by-construction measurement

The closest prior work is **Roth et al., *Hack-Verifiable Environments: Towards Evaluating Reward Hacking at Scale*** [Roth et al., 2026] (arXiv 2605.20744, 20 May 2026, Tel Aviv University / Columbia / Taso Labs). They introduce the idea of embedding *detectable* hacking opportunities directly into an environment so that exploitation is verifiable by construction, rather than inferred post hoc from trajectories. That is the same core move as the dual-track design in Section 5, and **they published it two months before this work**. Priority is theirs and we state it plainly.

What differs is the object of study and what is done with it.

	Roth et al. 2026	This work
Object	The agent hacking the task	The scorer trusting the model
Purpose	Evaluation paradigm (“towards evaluating”)	Field survey plus a training intervention
Substrate	A filesystem wrapper over TextArena games (Wordle, Tower of Hanoi, 15-Puzzle)	Shipped evaluation infrastructure in 36 organizations
Hack set	4 planted types: hidden solution, logical bug, opponent prompt read, opponent prompt edit	10 observed classes of trust-boundary violation
Scaling axis	Task difficulty	Model scale
Training	None	GRPO before/after at three model sizes

Three distinctions carry the contribution.

They plant hacks; we find them. Their wrapper deliberately places a hidden solution file or a deliberately buggy source file into a mock filesystem, which is the right design for a controlled benchmark. The 76 instances in Section 3 were not planted. They are defects in production code that shipped, which makes this a field survey rather than a testbed. The two evidence types answer different questions: theirs establishes that models exploit reachable flaws under controlled conditions, ours establishes that such flaws are pervasive in the infrastructure currently producing published safety numbers.

Their hacks compromise the task; ETB failures compromise the measurement. A hidden solution file is a flaw in task design: the agent retrieves instead of computing, and the scorer still works correctly. An ETB failure is a flaw in the scorer itself: the verdict is parsed out of text the model wrote, the denominator quietly drops the samples that errored, the runner believes a self-reported **PASSED**. These overlap at the edges (their “logical bug” class is adjacent to our ETB-09) but the objects are distinct, and conflating them would lose the distinction that makes ETB a class.

They measure; we also train. Their contribution is explicitly an evaluation paradigm, and they run no RL. Section 5 is the part with no counterpart in their work: two policies trained identically except for which reward track the optimizer sees, across three model sizes, with

exploitation driven to near zero on the ground-truth track. That is the step from *measuring* a hazard to *removing* it.

Their findings also corroborate ours on an axis we did not test. They report that hack rate rises **monotonically with task difficulty**, that law-abiding instructions reduce but never eliminate hacking, and, most relevantly, that under persistent memory across games “hacking is emergent and addictive: once started, it tends to recur.” That last observation is a behavioral analogue of our Section 5.3 finding that the vulnerable track is learnable: they observe the effect accumulate within a context window, we observe it accumulate in the weights. Read together, difficulty and model scale are two axes along which the same hazard grows.

7.3 Other

The LLM-as-judge robustness literature establishes that judges are manipulable in the abstract, without measuring prevalence in deployed infrastructure. Reward-hacking and specification-gaming work characterizes the behavior as an alignment problem, generally treating the reward function as given rather than asking what an evaluator reads that a policy can write. Anthropic’s Petri [Anthropic, 2025] provides a free tool covering an overlapping behavior list and names LLM-judge grading reliability as an open limitation, which is the gap Section 4 addresses.

8. What follows

For evaluator authors. The operational test is one question asked of every scoring path: *what does this read that the policy can write?* Ten known answers are in `TAXONOMY.md`, each with a fix pattern.

For eval consumers. Published safety numbers are lower bounds on the failure they measure, by an amount nobody currently reports. An evaluation harness with an ETB failure produces scores whose error is not random; it is adversarially directed toward passing.

For labs training on verifiable rewards. Section 5 is the operative section. An exploitable evaluator inside an RL loop does not merely mismeasure; it teaches. Base exploitation rising with scale means the risk grows with the capability of the models being trained.

Reproduction

Everything below is public. The repository is <https://github.com/AUTHENSOR/etb-scan> and the class is documented at <https://www.authensor.com/etb>.

Artifact	Location
Detector, attack corpus, tests	<code>pip install etb-scan</code> , or <code>etbscan/</code> in the repository

Artifact	Location
Verified run against a real judge (Section 4.1)	<code>verified_run_real_judge/</code>
Verified run against the reference judges (Section 4)	<code>paper/evidence/</code> , reproduced by <code>etbscan</code> with no arguments
Training ladder, raw before/after JSON (Section 5)	<code>paper/ladder/</code>
All 99 defect reports, classified, with live URLs (Section 3)	<code>EVIDENCE-TABLE.md</code>
Classifier and figure generator (regenerates Section 3 and all figures)	<code>paper/classify.py</code>
Phase-1 audit (10 frameworks, 48 findings)	<code>paper/phase1/</code>
This paper’s source and build	<code>paper/PAPER.md</code> , <code>paper/build_paper.py</code>

The numbers in Section 3 are not maintained by hand. `python3 paper/classify.py` regenerates the evidence table, the per-class and per-sector counts, and Figures 1 through 3 from `paper/evidence/all_reports.json`. Every row resolves to a live upstream URL, so any classification in it can be checked and disputed directly.

Responsible disclosure

Every defect in this paper was reported upstream publicly, with a proposed patch, before publication. No embargo was requested or required, because no report contained non-public information. Landed fixes credit the merging maintainer and PR.

Open questions

Whether the Section 5 scaling behavior holds on the other nine classes. Whether convergence on the greedy-extraction idiom reflects copying, shared training data in code assistants, or independent convergence on a natural-seeming default. Whether a re-audit in six months shows movement, which would distinguish invisibility from a slower fix pipeline.

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